LEARNER GUIDE

# Bounds of a rounded value

Percentages, Ratio, Growth and Bounds · Lesson 8. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

## What this lesson is about

A rounded value stands for every value that rounds to it. The smallest is the **lower bound**; the values stop rounding to it at the **upper bound**. Both are **half a unit** of the rounding away from the rounded value. The lower bound is included, but the upper bound is not, because it rounds up. Write the **error interval** as lower bound $\leq x <$ upper bound.

## What you are learning to notice

- First find the unit: 10 for the nearest 10, 0.1 for 1 decimal place, and so on.
- The bounds are half a unit either side of the rounded value.
- The lower bound is included (≤); the upper bound is not (<).
- Do not write the upper bound as 7.349 or 7.3499…: write 7.35.
- The same value has different bounds at different degrees of accuracy.

## Moving through it, one step at a time

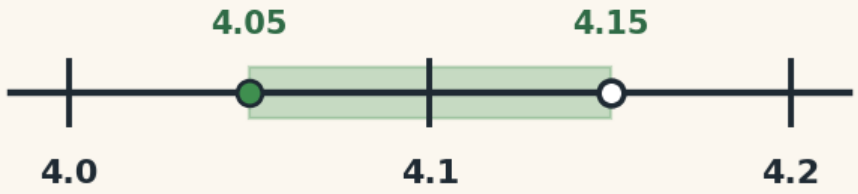
Say what the value was rounded to, and write down the unit. Halve the unit. Take it away for the lower bound and add it on for the upper bound. Write the error interval: lower bound $\leq x <$ upper bound. Check: the lower bound rounds to the value; the upper bound rounds up to the next value.

## Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

**Example 1 — To 1 decimal place**

A length is 4.1 m to 1 decimal place. Find its bounds and write its error interval.

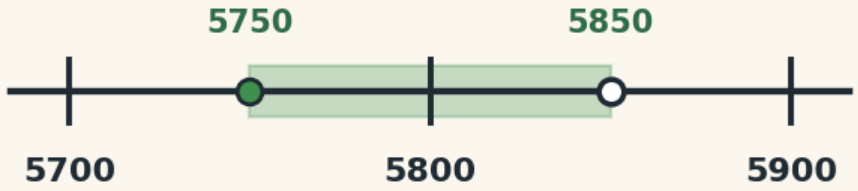



| | |
|---|---|
| **1.** The unit | 1 decimal place: the unit is 0.1 |
| **2.** Half a unit | Half of 0.1 is 0.05 |
| **3.** The bounds | 4.1 – 0.05 = 4.05;<br>4.1 + 0.05 = 4.15 |
| **4.** Check | 4.05 rounds to 4.1;<br>4.15 rounds up to 4.2 |

**$4.05 \leq x < 4.15$.**

**Example 2 — To 2 significant figures**

A town's population is 5800 to 2 significant figures. Find its bounds.




| | |
|---|---|
| **1.** The unit | The second significant figure (8) is in the hundreds: the unit is 100 |
| **2.** Half a unit | Half of 100 is 50 |
| **3.** The bounds | 5800 – 50 = 5750;<br>5800 + 50 = 5850 |

**$5750 \leq P < 5850$.**

**A gentle reminder**

You can pause. You can return. You do not need to complete everything in one sitting.

## If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (rounding to decimal places and significant figures), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

**How the worked solution unlocks**

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

## Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

## Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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